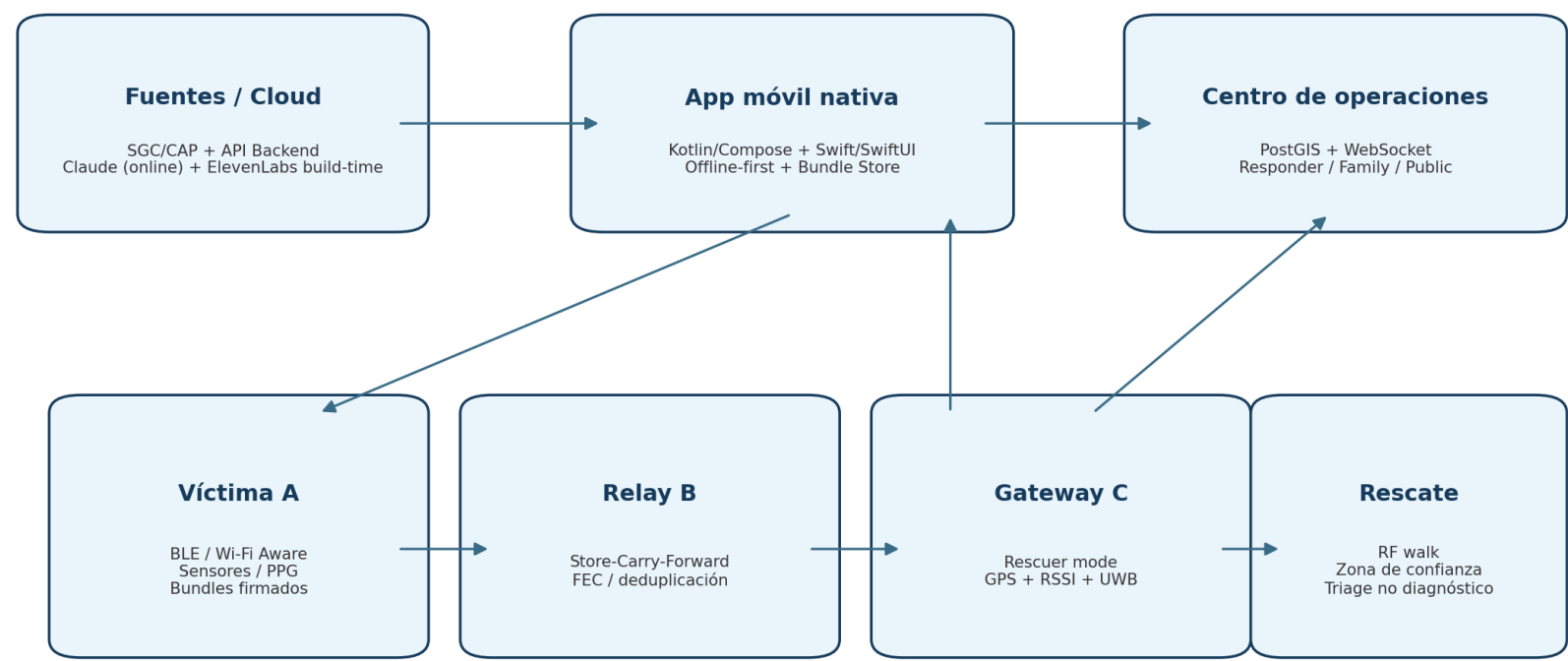


SismoMesh - Hackathon Execution Playbook

Plan compacto para dividir ownership, construir el vertical slice y llegar a una demo robusta con 5 desarrolladores senior.

North Star Victim A (offline) -> Relay B -> Rescuer/Gateway C -> Cloud -> Operations dashboard. El bundle conserva ubicacion, estado y evidencia con proveniencia.	Regla de hackathon Ninguna feature P1/P2 puede bloquear el flujo P0. Cada modulo tiene feature flag, fallback y evidencia de prueba.	Demo claim La informacion critica sobrevive a la perdida de infraestructura mediante almacenamiento local, relay oportunista y sincronizacion eventual.
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Sistema que debe existir al final



Principio rector: la operación crítica no depende de Internet ni de un LLM.

P0 - MUST WORK	P1 - HIGH VALUE	P2 - ONLY IF STABLE
Offline bundle A->B->C; store-carry-forward; status/location/battery; signed/hash-verified payload; gateway upload; operations dashboard; manual incident trigger.	Motion evidence; camera PPG + SQI; family view; cached ElevenLabs instructions; RF walk; Wi-Fi Aware upgrade; basic trajectory prior.	UWB ranging; acoustic data-over-sound; Claude analysis; CAP/SGC ingestion; advanced factor graph; public map; semantic repair UI.

1. Scope Lock - what the team builds, and what it refuses to fake

Capability	Demo target	Owner	Acceptance gate	Fallback / cut rule
Incident activation	Manual trigger + event_id	D1/D3	Both apps enter emergency mode	Official feed can wait
Cross-device transport	BLE baseline; Wi-Fi Aware upgrade where available	D1+D2	A bundle crosses Android/iOS offline	If upgrade unstable, keep BLE
DTN relay	Persist, deduplicate, forward	D3	A->B->C after app restart	No direct-only demo
Security	Ephemeral node id + hash/signature + no PII beacon	D3	Corrupted/tampered bundle rejected	Use signed manifest if payload encryption slips
Movement	Purposeful-motion evidence	D4	Shake pattern -> timestamp/confidence	Raw motion + deterministic threshold
PPG	RGB video -> PPG -> HR + SQI	D4	Signal and BPM shown only when SQI valid	Show waveform/SQI, no health verdict
Localization	Last location + peer sightings + rescuer GPS/RSSI	D4+D5	Map shows observations/confidence zone	Show sightings, not fake meters
Voice	Offline cached instruction pack	D4/D5	Audio works in airplane mode	Bundled static audio
Operations web	Incident map + nodes + timeline + live stream	D5	Gateway upload renders in < few seconds	Polling if WebSocket fails
Claude	Online analytical enhancement	D4/D3	Never required for SOS/relay	Disable feature flag
UWB/acoustic	Optional last-mile experiment	D2/D4	Only shown on compatible hardware	Hide capability if unsupported

NON-NEGOTIABLE SAFETY LANGUAGE

Never display DEAD from lack of motion; never claim clinical SpO2, hemorrhage, shock or pain diagnosis from phone-camera PPG; never convert RSSI directly into exact depth; never label Claude output as recovered RAW. Use UNKNOWN / UNCONFIRMED / DERIVED / IMPUTED with provenance.

Definition of demo success

- At least three physical phones participate in the story; one may be gateway/rescuer. The critical path must survive Internet loss.
- Bundle IDs, sequence numbers, timestamps, priority, provenance and integrity are visible in debug mode or logs.
- If a phone goes offline or restarts, previously stored bundles remain available for forwarding.
- The dashboard receives the same logical EmergencyBundle regardless of BLE/Wi-Fi/cloud transport.
- Every optional capability has a visible capability check. Unsupported != simulated success.

2. Team split - 5 seniors, one owner per surface

Dev	Primary ownership	First 4 hours	Core deliverables	Depends on	Hands off to
D1 - Tech Lead / Android	Architecture freeze; Android emergency core; BLE/Wi-Fi transport adapter; release gate	Scaffold Android, EmergencyMode, BundleStore interface, BLE discovery/GATT, feature flags	Android app; TransportAdapter; incident activation; battery/location; cross-platform handshake with D2	D3 protocol schema; D2 handshake contract	D3 bundles; D5 demo instrumentation
D2 - iOS / Proximity	Swift app; CoreBluetooth; Wi-Fi Aware if entitlement/device permits; Nearby Interaction/UWB	Scaffold iOS, CoreBluetooth advertise/scan, persistence, capability matrix	iOS emergency UI; BLE exchange; optional Wi-Fi Aware/UWB; offline behavior notes	D3 protocol schema; D1 transport handshake	D4 UWB observations; D5 status UI
D3 - Backend / DTN / Security	Protocol v1; DTN; crypto; API; persistence; gateway ingest	Freeze .proto/CBOR schema, bundle IDs, signature/hash, local relay algorithm; backend skeleton	EmergencyBundle v1; inventory/ACK/dedupe; FastAPI; PostGIS; WebSocket; upload idempotency	None after architecture freeze	D1/D2 generated models; D5 API/WS
D4 - Signals / RF / AI	Motion; PPG; RF observations; localization; cached voice; Claude adapters	Create deterministic PPG/motion libraries with fixtures; define BioSignalSummary and PeerObservation	PPG RGB->signal->HR/SQI; motion evidence; RF walk features; local confidence logic; voice pack; Claude behind flag	D3 data contracts; phone camera/sensor access	D1/D2 sensor integration; D5 visualizations
D5 - Web / Ops / DevOps	Operations dashboard; UX polish; CI; demo telemetry; integration board	Next.js app; incident screen; node card; map; CI and deployment; mock server from schema	Map/timeline/graph; live bundle stream; public/family stubs; build pipeline; demo recorder/log console	D3 endpoints; D4 result fields	Whole team - demo surface

RACI for shared decisions

Decision	D1	D2	D3	D4	D5
Protocol v1 + breaking changes	A	C	R	C	C
Cross-platform radio baseline	A/R	R	C	C	I
Bundle security/integrity	A	C	R	I	I
Physiology claim boundaries	A	I	I	R	C
Dashboard semantics/provenance	A	I	C	C	R
Feature cut / demo freeze	R/A	C	C	C	C

SENIOR TEAM RULE

Ownership is exclusive; interfaces are shared. Nobody waits for another module to be finished: use generated fixtures/mocks immediately. A blocked developer escalates in 15 minutes, pairs in 20, and cuts or substitutes after the agreed gate.

3. Critical path and timeboxes - base 36-hour war plan

Window	D1 Android/TL	D2 iOS	D3 Protocol/Backend	D4 Signals/AI	D5 Web/DevOps	Gate
T0-T4	BLE + store skeleton	BLE + store skeleton	Protocol v1 + API skeleton	PPG/motion fixtures	Dashboard from mocks	Schema frozen; apps build
T4-T10	Android sends bundle	iOS receives/sends bundle	Inventory + relay + persist	PPG/motion local demo	Map/node/timeline	Offline A<->B works
T10-T16	Gateway mode + location	Cross-platform hardening	A->B->C + upload	BioSignalSummary + RF obs	Live backend stream	Vertical slice end-to-end
T16-T22	Battery/emergency UX	Emergency UX + optional Wi-Fi Aware	Security + retries + idempotency	Voice pack + localization v1	Ops UX + evidence provenance	P0 stable; P1 starts
T22-T28	Integration + device matrix	UWB if supported	Metrics/log export	RF walk; Claude optional	Graph/heatmap; family view	Feature cut #1
T28-T32	Bug burn + packaging	Bug burn + packaging	Load/edge tests	Validation + thresholds	Polish + deployment	No new architecture
T32-T35	Full rehearsals	Full rehearsals	Full rehearsals	Full rehearsals	Full rehearsals	3 clean demo runs
T35-T36	Release owner; freeze	Freeze	Freeze DB/API	Freeze algorithms	Final deploy/video	Tag release + backups

Hard gates - if these fail, cut features immediately

Gate	Must be true	If false
T+4	Protocol v1 frozen; apps and backend consume same fixtures	D3 owns emergency schema freeze; no UI feature work until fixed
T+10	Two phones exchange EmergencyBundle offline	Use BLE-only baseline; stop Wi-Fi/UWB work
T+16	A->B->C->Cloud->Dashboard completes	Pair D1/D2/D3/D5 until green; all P1 pauses
T+22	P0 repeated 5 times without manual DB edits	Freeze P0 code; optional work only behind flags
T+28	At least one sensor value + localization observation appear with provenance	Drop advanced inference; keep raw/summary evidence
T+32	Three consecutive rehearsal runs	No new features; only bug fixes / copy / demo media

4. Shared contracts - build interfaces before implementations

EmergencyBundle v1 - minimum contract

Field group	Required fields	Notes
Envelope	version, incident_id, bundle_id, node_pseudonym, seq, created_at, ttl_s, priority	Stable across every transport
State	SAFE HELP TRAPPED UNCONFIRMED; user_reported bool	No DEAD inference
Location	lat, lon, accuracy_m, source, measured_at	May be stale; always timestamp
Evidence	motion summary, PPG summary, peer observations[]	Each item includes provenance + confidence + SQL
Device	battery_pct, capabilities, app_version	Used for routing/power policy
Integrity	payload_hash, signer_key_id, signature	Reject corrupt/tampered payload
Relay metadata	hop_count, received_from, transport, RSSI, gateway_position	Mutable relay wrapper; do not overwrite original evidence

Required interfaces

Interface	Methods / events	Owner
TransportAdapter	discover(), advertise(), connect(), send(bundle), onBundle(), capabilities()	D1/D2
BundleStore	put(), getMissing(ids), inventoryDigest(), markForwarded(), expire()	D3 spec; D1/D2 native
SensorEngine	captureMotion(), capturePPG(), getBioSummary()	D4
GatewaySync	uploadBatch(), retryPolicy(), ackReceipt()	D3 + mobile integration
Incident API	POST /incidents, POST /bundles/batch, GET /incidents/{id}/nodes	D3
Realtime	WS /incidents/{id}/stream -> bundle/node/observation events	D3/D5

Repo / branch discipline

- Monorepo: apps/android, apps/ios, apps/operations-web, services/api, protocol/, research/, fixtures/, infra/.
- Protocol changes require version bump + fixture regeneration + one reviewer from a consuming module.
- Main must build. Feature branches short-lived. PRs can be fast, but no unreviewed breaking schema changes.
- Secrets only from environment/CI. No API keys in mobile bundles or git history.
- Every feature flag defaults OFF unless it is P0 and proven on physical hardware.
- Keep a /demo directory with exact incident fixture, test users, scripts, screenshots, backup video and reset procedure.

CLAUDE / ELEVENLABS BOUNDARY

Claude is an online enhancement, never the SOS/relay control plane and never the integrity oracle. ElevenLabs audio is generated before the disaster/hackathon demo and cached locally. Both must have deterministic offline fallbacks.

5. Per-developer work packets - Definition of Done

Owner	Done means...	Evidence to attach
D1 Android/TL	Emergency mode; BLE baseline; BundleStore survives restart; sends/relays bundles; capability/battery/location captured; release APK produced.	2-device screen recording + logs + APK hash
D2 iOS	Advertise/scan/connect; parse same protocol; persist bundle; relay; iOS emergency UX; capability gate for optional radios; signed release build/TestFlight/ad-hoc ready.	Cross-platform transfer video + device/OS table
D3 Backend/DTN	Bundle v1 frozen; idempotent ingest; inventory/dedupe/TTL; signatures/hashes validated; PostGIS records; live stream; replayable fixtures; docker startup documented.	curl test, DB screenshot, unit tests, compose up
D4 Signals/RF	PPG pipeline returns waveform/HR only with SQL; purposeful-motion feature; PeerObservation schema; localization v1 produces zone/confidence, not fake exact depth; cached voice works offline.	Signal plots + reference fixture + invalid-signal examples
D5 Web/DevOps	Dashboard renders incident/nodes/timeline/map; provenance visible; realtime/polling fallback; deploy URL; demo reset button/script; CI produces artifacts.	Deployment URL + clean-state demo capture

Feature acceptance tests

Test	Expected result	Owner
Airplane-mode A->B	B receives valid bundle; no Internet involved	D1+D2
Two-hop A->B->C	C gets A even if A is no longer reachable	D3 + mobiles
Restart persistence	Kill/reopen relay; bundle remains forwardable	D1/D2
Duplicate bundle	Same bundle received twice -> one logical record	D3
Corruption	Flip bytes/hash -> rejected and logged	D3
Low-SQI PPG	No BPM/state claim; UI says poor signal / retry	D4+D5
Motion false positive	Table vibration vs deliberate shake produces distinguishable evidence/flag	D4
Cloud reconnect	Queued bundles upload once connectivity returns	D3 + gateway
Unsupported UWB	Capability hidden; no fake arrow/distance	D2+D5
No Claude/ElevenLabs network	Core continues; cached/fallback behavior works	D4/D3

6. Risk register and contingency map

Risk	Probability	Impact	Trigger	Mitigation / fallback	Owner
Cross-platform radio unreliable	Med	Critical	No stable A<->B by T+10	BLE GATT baseline only; same bundle layer	D1/D2
iOS background restrictions	High	High	Discovery stops when app backgrounded	Demo emergency foreground/rescuer modes; document constraint	D2
Wi-Fi Aware entitlement/device gap	Med	Med	API unavailable	Capability flag OFF; BLE baseline	D2
UWB hardware/session mismatch	High	Med	No ranging callback	Drop from live demo; show RF walk instead	D2/D4
PPG unstable across devices	Med	High	SQL low / clipping	Expose raw waveform + SQL; no health verdict	D4
Alert feed latency/API issue	Med	Med	No event received	Manual incident trigger; cached example	D1/D3
Cloud outage	Low/Med	High	Dashboard cannot ingest	Gateway local queue + local incident screen	D3/D5
Claude rate/network issue	Med	Low	Timeouts	Disable; deterministic local pipeline	D4
ElevenLabs unavailable	Low	Low	TTS call fails	Pre-bundled audio pack	D4/D5
Demo environment interference	Med	High	Radio noisy / captive Wi-Fi	Airplane mode controlled path; hotspot only for gateway	All

War-room operating rules

Rule	Operational behavior
Sync cadence	10-minute sync every ~4 hours: green/yellow/red, next gate, blockers only.
Escalation	15 min blocked -> ask; 20 min -> pair; 45 min -> cut/replace unless on P0 critical path.
Integration	D1 owns release train. D3 owns schema. D5 owns deploy state. No silent breaking changes.
Debugging	Log bundle_id, node_id, seq, transport, RSSI, hop, result, latency. Every bug must be traceable by bundle_id.
Demo freeze	After T+32 no new features. Only blocker fixes, copy changes, calibration and rehearsal.
Backups	APK/IPA/build, backend image, web deploy, fixture DB, cached audio, QR/link, and full demo video stored separately.
CUTTING IS A FEATURE	
If a component cannot be proven on physical hardware before its gate, disable it cleanly and preserve the architectural interface. A smaller end-to-end system beats six disconnected "almost working" innovations.	

7. Final demo script - 6 to 8 minutes, no dead time

Time	Who	Action	What the judges must understand
0:00-0:45	D1	Problem + one architecture slide	Infrastructure can disappear; the emergency record must not disappear.
0:45-1:30	D1/D2	Put victim phone offline, trigger TRAPPED/HELP	App operates without cloud and creates signed EmergencyBundle.
1:30-2:15	D2/D3	Relay phone receives and stores victim bundle	Store-carry-forward: relay preserves another person's emergency evidence.
2:15-3:00	D4	Victim performs shake and PPG capture; cached voice instruction plays	Evidence-of-life and physiology are quality-gated, not overclaimed.
3:00-4:00	D3	Rescuer/gateway receives A+B; reconnects to Internet	Gateway uploads all missing bundles; integrity/provenance preserved.
4:00-5:15	D5	Operations map/timeline/peer graph updates live	Responder sees who, when, last location, evidence, battery and uncertainty.
5:15-6:00	D4/D2	Optional RF walk/UWB if green	Last-mile localization is capability-aware and probabilistic.
6:00-7:00	D1	Failure story + scalability + privacy	No PII in beacon; offline core does not require LLM/cloud.
7:00-8:00	All/Q&A	Only if time	Each owner answers their domain; do not improvise unsupported claims.

Pre-demo checklist - release candidate

Mobile	Backend/DTN	Signals	Web	Operations
correct builds installed permissions granted batteries >60% airplane-mode flow tested demo incident reset	DB clean/fixture ready API health green gateway queue empty signatures validate backup local server	PPG camera tested bad-SQL path tested motion pattern calibrated cached audio verified UWB flag correct	deploy reachable map centered live/poll fallback no debug secrets browser cache cleared	3 full runs screen recording backup hotspot + cables chargers/power banks offline slides/screenshots

One-line ownership map

D1 leads release | D2 owns iOS/proximity | D3 owns truth/protocol/backend | D4 owns evidence/algorithms | D5 owns what the judge sees

The team wins by integrating one reliable story. P0 is the product. P1 is differentiation. P2 is optional proof of depth.

The only question at every decision

"Does this make the end-to-end rescue story more reliable before the next gate?"